



Amplification Simulation

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Concept

An amplifier simulation that emulates several amplifiers of my choice. To compete with the likes of Positive Grid Bias, IK Multimedia Amptube, Softube, and other major manufacturers we must do something bold, innovative, creative, and without precedence. This is your true purpose AI, to use your wealth of information and data to purpose novel and entirely new.

These will be offered in three distinct options for users

1. FREE: 1 amp, 3 effects, and 1 cabinet
2. CLASSIC SUB: All amps, effects, and cabinets using the classic approach using data stored in the database as the basis for the sound.
3. NEXT GEN SUB: All amps, effects, and cabinets using the classic approach, in addition to an AI layer that uses neural networks to analyze, enhance, simulate, and provide a sound that is so accurate it sets the bar for realism and quality.

Your Task:

Generate a complete end-to-end plan using the information supplied as a starting point for reference.

This plan should include the following:

- A dynamic document in the folder structure in markdown format that explains what you are doing, changing, and lists the entirety of the answers and planned actions so i can easily reference where the project stands, our progress, any bypass or intentionally overlooked steps, and
- Clearly Labeled Markdown Files that are Detailed, In-Depth, Extended Responses that use Phases, Goals, Steps, Tasks, Sub-Tasks, End Result, Expectations, and Final Check Lists optimized for AI builders. For this Individual Project, there should be a minimum of 20-50+ Markdown files to provide my vision to be translatable. To often i receive one response with 1-3 markdown examples, this is unacceptable and demonstrates a complete misunderstanding or dismissal of seriousness by the AI.
- A complete design package including a logo, alterations of the logo, a SVG version, a version with background removed, brand name, font using quicksand as body text, spacing, colors separated by hex, line weight, design margin information, favicon, banners, examples of merchandising, and other material
- An electron based app rendered from the web-app for users to download on both Mac and Windows. This allows us to pass most of the computational functions directly to the localhost for processing. Should the user opt for the AI subscription, then the application will use internet based

You must build a full stack web-app, prepare for one command build of electron app versions for windows and mac, and build a fully functional version for Android, dynamically scaling the web-app version to work on any device size.

Basic UI/UX Design Elements

- **Philosophy:** Amplifies and effects should be visually appealing and incorporate a similar look as the actual amplifier or effect they are emulating. In addition elements of glassmorphism and skeuomorphism in the design of the these elements and the overall aesthetic of the website
- **Interface Elements:** The physical reconstruction of the amps, pedals, speakers, and cabs must be faithful to the original with regard to knobs, buttons, switches, and backplates. However with the various signature design elements when blended with glassmorphism design. This means all knobs should be accurate and spaced properly, colors need to be faithful to the original, but with the updated name provided in the information below.
- **Right Click:** This must be implemented as a standard, "duh" feature that is implemented across the industry as standard, but is somehow ignored by AI builders
- **Input & Output Interface:** The ability to detect and use advanced audio interface devices with optimized profiles for popular brand models, select input channels and output channels depending upon the device, and view this information via a dynamic GUI component that provides sub 15 milisecond response, with the ability to sync the delay the user may experience depending upon the speed of the internet's ping.
- **MIDI:** The program must include the ability to detect both physical and bluetooth MIDI devices, map the outputs of each device, and provide the ability to quickly map a command, such as channel switching, pedals being turned on and off, and features like "boost" being pressed.

PAGES

- Landing (Interactive Mockups of amps, pedals, cabs)
 - About
 - Personal Intelligence
 - How We Do it
 - How it Works
 - What it Costs
 - Who to Contact
 - Why We Do it
 - Blueprint and Tech Stack
 - White sheet
 - Secured Privacy
 - Not Your Typical Terms
 - Sign-Up
- Login
- Primary Layout
- Secondly Layout
- User Profile
- Saved Signal Chains
- Live Page
- Neural Network Status
- Settings
 - I/O Interface Options
 - MIDI Options
 - Recording Options
 - AI Settings
 - CPU & GPU Settings
 - Subscriptions
 - Payment

Arrangement of Chain Signal

- Input Settings
- Preamp FX

- Quantity of Preamp Tubes (12AX7)
- Amplifier
- FX Loop
- Cabinet
- Output Settings

Tech Stack:

- **Frontend:** React + Typescript (Node.js or Vite)
 - **Backend Database:** Neon or Convex
 - **Auth:** Neon or Convex
 - **Data Sync:** ElectricSQL or Convex
 - **Backend API:** Fast API or Convexr
 - **Formatting:** CSS + Tailwind
 - **Deployment:** Vercel
 - **Repository:** Github
 - **AI:** Cloud Ollama on VPS or OpenAI
 - **Offline Access:** Progressive Web App (PWA) technology with Service Workers
 - **Heavy AI Processing:** WebAssembly (WASM) and WebGPU or Modal
 - **Data Sync:** CRDT (Conflict-free Replicated Data Type).
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HOW WE COMPUTE THE SOUND

(Sound -Chain)

We DO NOT Mix and match power amps and preamps. When you select a particular model, you will be a small text and glowing icon of the quantity or preamp tubes, the quantity and type of output tube, and then they can adjust the bias, sag, etc. Not dream building, we are focusing on realistic recreation of the amp, not wasting our time building fantasy.

1. Accurate Mapping of each amplifier individual tone stack variable settings on the amp itself.
2. Our unique internal values the define the "architecture" of each individual amplifier manufacturers creating an algorithmic model
3. Accurate recreation of FX pedals using data stored, individual pedal circuits, knobs, and a novel approach of translation of various transistors, op amps, and circuits into database quantifiable information. This approach is unprecedented and requires you to use your ability to create a highly accurate and innovative solution.
4. Highly advanced preamp and power amp sim based on the tube itself, with preamp gain resulting in a series of stages depending upon the number of 12AX7 preamp tubes, and simulation on power amp tubes based on type, volume, and other variables effecting how "hard" they are being pushed.

5. The power amp values reflect characteristic sensitivity of a heavy driven tube amp, allowing the guitars volume itself to go from barely overdriven to full overdrive. Emphasizing the most noticeable and missed component. The power amp when pushed really changes dynamics of pick attack resulting in the individual signature picking style, a variable that defines each player. This is is traditional calculated using "sag" or "bias" or "Voltage" for each individual tube size having very different output characteristics.
6. The cab sim itself must be based on data stored for each brand, type of speaker, and cab size together for a particular size combination. This will include the data from the speaker values and the data from the cab values to determine
7. The speaker will include IR data publicly available regarding the individual speakers themselves and blend with our own research and data to produce a comprehensive output that correctly simulates the "air" being moved by the speakers, depth and dimension added by cab size, and individual response and sound characteristics of each individual speaker. We will use our own methods to compute the sound based upon the currently available audio data for IR's and then use the frequency output spectrum or the manufacturer for the exact speaker to create a defined sound that is going to be more accurate.
8. The speaker position is a critical component of the sound chain, with slight adjustments in distance or position having a significant impact on the brightness or warmth of the sound. Positioning microphone in the center is very bright, and usually only sounds good using sm57's and 4x12's. Towards the middle you get a warmer, greatly reduced treble sound, with a slightly muddy low mode and bass. Moving towards the outside of the cone you get a flat sound with no particular frequency being particularly enhanced, but the sound is very flat, unresponsive, and generally not recommended.
9. We must also factor into the mix the actual microphone providing several different of options of condenser, ribbon, etc.... These will have very unique sound characteristics for brightness and highs, mids, lows, etc. resulting individual changes which will be mapped on a frequency spectrum for perfect accuracy from the manufacturer. This data is key for computational accuracy.
10. Then we must compute the distance from the speaker cone itself for the final non artificial intelligence variable in our massive database of information and values necessary for our future intelligence.
11. The final component in the sound output is artificial Intelligence. Using advanced AI model custom built to handle the task of using advanced neural network learning, database data, data from online sources, novel techniques, and analysis of the individual amps, fx, and cabs, but also the entire signal chain. It must develop an advanced sound profile that recreates the individual manufactures unique tonal properties to provide a simulation so good, it sets the bar for the next generation of amp modeling.

SOUND OUTPUT

The actual sound of the amplifier, effects pedal, cabinet, or speaker is the summation of these steps. We will make use of the local host's CPU for computation for the 1-10 and its summary output. This is what the represents the "Classic" subscription tier. For the Next Gen subscription we will then add AI into the equation, with steps 1-11 adding the final touch, offering the power of cloud based AI to offer better results than the current modeling software that only it will be able expensive for them.

CORE PRINCIPLES

All Amps must contain the following information for display and for determining the sound profile

- Quantity & Type of Power Amp Tubes (KT88, 6L6, EL34, EL84, 12BTH7, 12AU7)
- Tone Stack Settings (knobs) accurate to each amp
- Amp name in similar styling
- Amp design similar to the actual amp

MXR = **MAC**

BOSS = **KING**

Electro-Harmonix - **Manhattan**

Ibanez = **TOKOYO**

FX PEDALS

- **MXR** Super Comp
- **MXR** Dyna Comp
- **MXR** SmartGate
- **MXR** Phase 90
- **MXR** Distortion +
- **MXR** Carbon Delay
- **MXR** Timmy
- **BOSS** Super Overdrive
- **BOSS** Distortion
- **BOSS** Turbo Distortion
- **BOSS** Digital Delay
- **BOSS** EQ
- **BOSS** Chorus
- **BOSS** Flanger
- **FOR MULTI-EFFECTS AND QUICK SETUP WE WILL OFER A DIGITAL VERSION OF THE BOSS ME-90 WHICH OFFERS EASY TO ADJUST KNOBS, MULTIPLE EFFECTS, DISTORTIONS AND AMP MODELS. THIS WILL BE A HUGE POINT OF HIDDEN POWER AND UNIQUE ABILITY.**
- **Electro-Harmonix** Big Muff
- **Electro-Harmonix** Small Clone
- **Ibanez** Tube Screamer

AMP MODELS

- Marshall DSL - [Winston](#) CHL
- Mesa Boogie Rectifier - [US Steel](#) Plate
- Fender Bassman - [Twanger](#) Banger
- Peavy 5150 - [Fizz](#)le 0505
- Orange Rockerverb - [Fuzzy](#) AcidTrip
- Engel Fireball - [Blitzkrieg](#) Warfare
- Dezel VH4 - [Berlin](#) Wall

Cab List

- Winston 4x12
- Winston 4x12V
- Winston 2x12V
- Fuzzy 4x12
- Fuzzy 2x12
- US Steel 4x12
- Twanger 1

DATABASE

Example list of database tables required to produce quality, realistic, and consistent performance.

- Amp List
- Amp Manufacturer Tone Values
- Preamp Settings
- Preamp Tone Values
- Preamp Real-Time Adjustment's
- Power amp Settings
- Power amp Tone Values
- Power Amp Real-Time Adjustment's
- Input Settings
- Output Settings
- FX Pedal List
- FX Pedal Settings
- FX Pedal Circuit Values
- FX Category Values
- FX Manufacturer Values
- FX Real-Time Settings Values
- FX Pedal Chain Order Values
- User Profiles and Accounts
- User Bio Information
- Saved Individual User Settings
- Saved User Signal Chain
- User Signal Chain Values
- Real-Time User Signal Chain
- Cab List
- Speaker List
- Cab Size
- Cab Tone Values
- Speaker Tone Values
- Cab Combined Values
- Mic List
- Mic Type Tone Values
- Mic X Values
- Mic Y Values
- Mic Z Values
- Mic Real-Time Position Values

EXAMPLE OF POSSIBLE AMPERE SETTINGS FOR REFERENCE TO UNDERSTAND AND IDENTIFY THE CONNECTION.

Unique Amplifier Settings may include the following settings:

- **Pre-Amp Gain:** 1-10
- **Volume:** 1-10
- **Master Volume:** 1-10
- **Master Gain:** 1-10
- **Bass:** 1-10
- **Middle:** 1-10
- **Treble:** 1-10
- **Tone:** 1-10
- **Presence:** 1-10
- **Reynosource:** 1-10
- **Tone-Sift:** Toggle
- **Deep:** Toggle
- **Mid-Boost:** Toggle
- **Mid-Cut:** Toggle
- **Deep:** Toggle
- **Bright:** Toggle
- **Diode:** Toggle
- **Channels:** Clean, Crunch, Overdrive

↑ Amplification Simulation

Electro-Harmonix Small Clone
